PharmaLeaf · B.Pharm Semester VIII · PCI 2026 (NEP 2020)

BP801T Ethical Considerations and Translational Applications of AI in Pharmacy

PharmaLeaf notes for B.Pharm Semester VIII (PCI 2026, NEP 2020). Written in easy language, unit by unit in syllabus order. Worked examples were computed in Python (scikit-learn) on small simulated teaching datasets; regulatory facts were checked against current FDA, EMA, CDSCO and Government of India sources (see Sources).

Syllabus at a glance

BP801T is a 2-credit core theory course: 2 lectures per week, 30 hours in total, 50 marks (20 sessional + 30 end-semester).

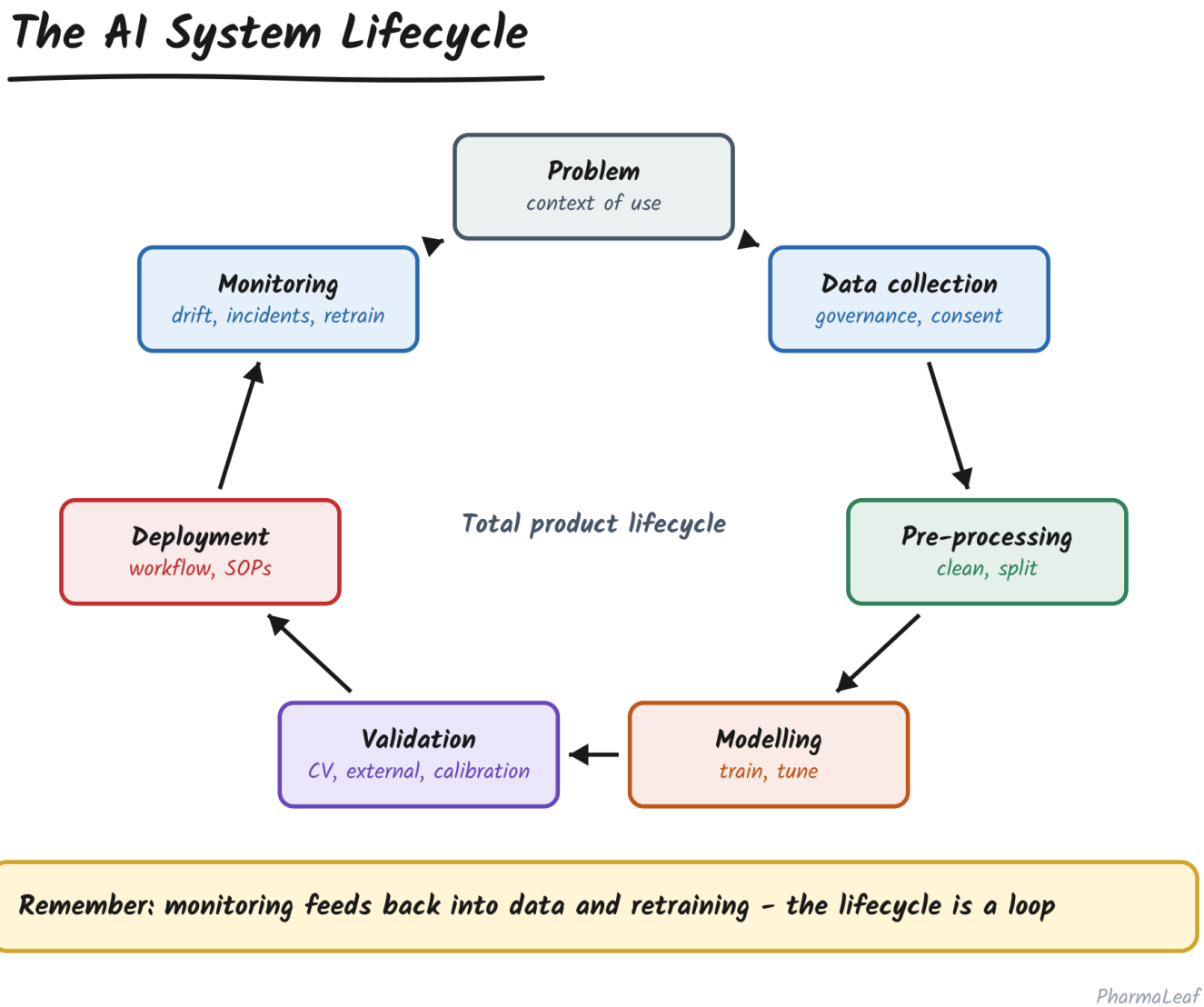
Unit	Title	Topics	Hours
I	AI lifecycle, validation and model auditing	Lifecycle stages; data quality; training, validation and test sets; cross-validation; model drift; data leakage; documentation and reproducibility; model auditing	6
II	Regulatory framework and explainable AI	AI in regulatory submissions; XAI; transparency and interpretability; EU AI Act, FDA and CDSCO frameworks; accountability; risk-based classification	6
III	AI in pharmacy automation and supply chain	Automated dispensing; inventory prediction; demand forecasting methods; adherence monitoring; supply chain risk prediction; predictive logistics; advantages, limitations, legal and privacy issues	6
IV	AI in public health and real-world data analytics	RWD sources; outbreak prediction; population risk modelling; regression in epidemiology; vaccination forecasting; case studies	6
V	Guided project: translational AI in pharmacy	Supervised ML model on pharmacy data; validation; regulatory and ethical analysis; implementation plan	6

***Translational AI** means moving an AI model **from a research idea to real, safe, routine use** in practice (a pharmacy, hospital, factory or public-health programme), which needs*

validation, regulation, ethics, workflow integration and monitoring, not just a good accuracy score.

Unit I: AI Lifecycle, Validation and Model Auditing (6 hours)

1.1 Overview of the AI system lifecycle



Stage	Activities	Pharmacy example (QC batch failure prediction)
1. Problem definition	Define the **question, users, context of use, success criteria**, risks	Predict which tablet batches will fail dissolution so QC can test them first
2. Data collection	Identify sources, permissions and consent, data governance	Batch records: granule moisture, hardness, drying temperature, QC results

Stage	Activities	Pharmacy example (QC batch failure prediction)
3. Pre-processing	**Cleaning**, missing values, outliers, units, encoding, scaling, **splitting**	Remove duplicate batches; impute missing moisture; standardise
4. Modelling (training)	Choose algorithm, train, tune hyperparameters	Logistic regression, random forest
5. Validation	**Internal (cross-validation)** and **external** testing; performance, **calibration**, fairness	5-fold CV, test set, new-site data
6. Deployment	Integrate into workflow (software, dashboard, LIMS / EHR), user training, SOPs	Risk score displayed in the QC system
7. Monitoring and maintenance	Track performance, **drift**, incidents; **retrain** under change control; eventually **retire**	Monthly check of predicted vs observed failure rates

Key idea: the lifecycle is a **loop**: monitoring feeds back into data collection and retraining. Regulators (FDA, EMA, CDSCO) expect a **total product lifecycle (TPLC)** approach with documentation at every stage.

1.2 Importance of data quality in healthcare datasets

"**Garbage in, garbage out**": a model can only be as good as its data. Healthcare data are often **messy, incomplete, biased and collected for other purposes** (billing, routine care).

Data quality dimension	Meaning	Healthcare problem example
Accuracy	Values are correct	Wrong units (mg vs mcg), typing errors in lab values
Completeness	Required data present	Missing weight or creatinine in 30% of records
Consistency	Same data agree across systems	Different drug names / codes in pharmacy and EHR
Timeliness	Data up to date	Old allergy lists
Validity / conformity	Values follow formats and ranges	Age 250 years; dates in the future
Uniqueness	No duplicates	Same patient registered twice
Representativeness	Data reflect the target population	Only urban, tertiary-hospital patients